

[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

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Class 3 Maths — CA-1 (Hard)

Class : 3

Subject : Maths (Math-Mela)

Max Marks : 70

Duration : 2 Hr 30 Min

Syllabus: Chapters 1–3 — What's in a Name?, Toy Joy, Double Century

General Instructions: All questions are compulsory. For Section A, write the correct option letter.

SECTION A — Multiple Choice Questions (1 Mark each)

- A basket has 187 mangoes. If they are packed into boxes of 10, how many full boxes can be made and how many mangoes are left over? (1 Mark)
A. 17 boxes, 7 left over
B. 18 boxes, 7 left over
C. 18 boxes, 0 left over
D. 19 boxes, 7 left over
- Which shape, when cut straight through the middle parallel to its base, always shows a circle as the cut surface? (1 Mark)
A. Cube
B. Cuboid
C. Cylinder
D. None of these
- A cuboid has length, breadth, and height all different from each other. How many pairs of identical (equal) faces does it have? (1 Mark)
A. 1
B. 2
C. 3
D. 6
- If the number 1_5 (tens digit missing) is greater than 143 but less than 168, which digit could go in the blank? (1 Mark)
A. 3
B. 5
C. 2
D. 9
- What is the greatest 3-digit number you can make that is less than 200 using the digits 1, 9, and 8 (using each digit exactly once)? (1 Mark)
A. 198
B. 189
C. 918
D. 981
- A toy is made by joining a cone on top of a cylinder. How many total flat faces does the combined toy have (counting only the outer visible flat faces)? (1 Mark)
A. 1
B. 2
C. 3
D. 4
- Which of these statements is TRUE about all cubes? (1 Mark)
A. A cube is never also a cuboid
B. Every cube is a special type of cuboid
C. A cube has curved surfaces
D. A cube has only 4 faces
- The sum of the hundreds digit and tens digit of a number is 3, and the number is between 100 and 200. Which of these could the number be? (1 Mark)
A. 121
B. 300
C. 213
D. 450
- If you remove one flat face from a cube by cutting it open flat (unfolding), how many flat pieces (faces) would you see in total in the unfolded net? (1 Mark)
A. 4
B. 5
C. 6
D. 8

SECTION B — Short Answer Questions (2 Marks each)

- A pond has 176 fish. If 38 fish are caught and 22 new fish are added, how many fish are in the pond now? (0 Marks)
- Explain why a sphere cannot be stacked like a cube, using the idea of flat and curved surfaces. (0 Marks)
- Write a 3-digit number where the hundreds digit is double the tens digit, and the ones digit is 5. Explain your reasoning. (0 Marks)
- Two toy blocks, a cube and a cuboid, look similar from one side. Explain one way you could tell them apart just by looking at all their faces. (0 Marks)
- A number lies between 150 and 200. Its tens digit is 7 and its ones digit is the same as its tens digit. What is the number? (0 Marks)
- A rectangular toy box (cuboid) is placed so its largest face touches the ground. Explain why this makes the box more stable than standing it on its smallest face. (0 Marks)
- If 3 groups of 100 marbles and 4 groups of 10 marbles and 6 single marbles are combined, what is the total, and which chapter concept does this use? (0 Marks)

8. A cylindrical toy roller and a spherical ball are both placed on a slope. Explain if there would be any difference in how they roll, based on their shapes. (0 Marks)
9. Write the number that is 25 less than 200, and explain the subtraction steps. (0 Marks)
10. A shape has 5 faces, 8 edges, and 5 corners, with one face being a square and the rest being triangles. What could this shape be, based on shapes discussed in class? (0 Marks)
11. Explain why comparing only the first digit of two 3-digit numbers is not always enough to know which is greater. Give an example. (0 Marks)
12. A toy maker has 3 cubes, 2 cuboids, and 4 cylinders. If each shape has a different number of flat faces, find the total number of flat faces among all these toys (spheres and curved-only surfaces do not count). (0 Marks)
13. A number between 100 and 200 has digits that add up to 10, and its tens digit is twice its ones digit. Find the number. (0 Marks)
14. Explain how you would check if a given box-shaped object is a perfect cube without using a ruler, just by observation. (0 Marks)

SECTION C — Short Answer Questions (Explain) (3 Marks each)

1. A trader has 200 oranges. Every day, the trader sells some oranges and this pattern continues: Day 1 sells 34, Day 2 sells 12 more than Day 1, and Day 3 sells 8 fewer than Day 2. Find the total oranges sold over the three days and the oranges left after Day 3. (0 Marks)
2. Explain, using properties of faces, edges and corners, how you would prove to a younger student that a cube and a cuboid are related but not exactly the same shape. (0 Marks)
3. Four numbers are hidden in a pattern: each number is 15 more than the previous one, starting from 105. Write the first four numbers in this pattern and explain if any of them cross into "Double Century" territory (200 or beyond). (0 Marks)
4. A model house is built using 1 cuboid (walls), 1 cone-topped structure (roof shaped like a cone on a cylinder tower), and cube-shaped decorative blocks. Explain how the different properties of these shapes (stability, rolling, pointed tops) make them suitable for their specific parts of the house. (0 Marks)
5. A number sequence begins at 200 and decreases by 13 each time: 200, 187, 174, Continue the sequence for two more terms and explain the pattern of subtraction used. (0 Marks)
6. Explain, using examples, why understanding 3D shapes is useful beyond the maths classroom - for example, in building, packing, or everyday objects. (0 Marks)

SECTION D — Long Answer Questions (5 Marks each)

1. A large box contains 200 building blocks made up of cubes, cuboids, cylinders, cones, and spheres. If there are 40 cubes, 35 cuboids, 45 cylinders, and 30 cones, find how many spheres are in the box, and then explain one property that makes spheres different from all the other four shapes in this box. (0 Marks)
2. A number pattern game: start at 100, and each turn you can either add 15 or subtract 8. After 3 turns of adding 15 each time, what number do you reach? Then, if you subtract 8 twice from that number, what is the final number? Show every step clearly and check whether the final number is still within the "Double Century" range (up to 200). (0 Marks)
3. Design a toy shop inventory list with exactly 200 toys total, using at least four different 3D shapes (cube, cuboid, cylinder, cone, sphere), showing how many of each shape you would stock and why, then verify your numbers add up to exactly 200. (0 Marks)